EXERCISE 7.6

1. Find the absolute extrema of the following functions on the given closed interval.

(i) $f(x) = x^2 - 12x + 10$; $[1, 7]$ (ii) $f(x) = 3x^4 - 4x^3$; $[-1, 2]$

(iii) $f(x) = 6x^{\frac{4}{3}} - 3x^{\frac{1}{3}}$; $[-1, 1]$ (iv) $f(x) = 2\cos x + \sin 2x$; $\left[0, \frac{\pi}{2}\right]$

2. Find the intervals of monotonicities and hence find the local extremum for the following functions:

(i) $f(x) = 2x^3 + 3x^2 - 12x$ (ii) $f(x) = \frac{x}{x-5}$

(iii) $f(x) = \frac{e^x}{1-e^x}$ (iv) $f(x) = \frac{x^3}{3} - \log x$

(v) $f(x) = \sin x \cos x + 5,\ x \in (0, 2\pi)$

OFWQ9

7.7 Applications of Second Derivative

Second derivative of a function is used to determine the concavity, convexity, the points of inflection, and local extrema of functions.

7.7.1 Concavity, Convexity, and Points of Inflection

A graph is said to be concave down (convex up) at a point if the tangent line lies above the graph in the vicinity of the point. It is said to be concave up (convex down) at a point if the tangent line to the graph at that point lies below the graph in the vicinity of the point. This may be easily observed from the adjoining graph.

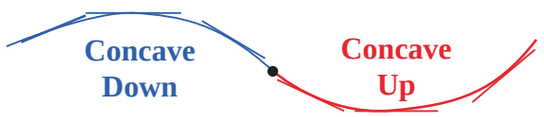


Fig.7.23

Definition 7.8

Let $f(x)$ be a function whose second derivative exists in an open interval $I = (a, b)$. Then the function $f(x)$ is said to be

(i) If $f'(x)$ is strictly increasing on I, then the function is concave up on an open interval I.

(ii) If $f'(x)$ is strictly decreasing on I, then the function is concave down on an open interval I.

Analytically, given a differentiable function whose graph $y = f(x)$, then the concavity is given by the following result.

Theorem 7.11 (Test of Concavity)

(i) If $f''(x) > 0$ on an open interval I, then $f(x)$ is concave up on I.

(ii) If $f''(x) < 0$ on an open interval I, then $f(x)$ is concave down on I.

Remark

(1) Any local maximum of a convex upward function defined on the interval $[a, b]$ is also its absolute maximum on this interval.

(2) Any local minimum of a convex downward function defined on the interval $[a, b]$ is also its absolute minimum on this interval.

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